Three Fixes to the AR(2) Extension of the Space-time Skew-t Model:
Why, How, and What the Tests Show

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1 Background

The Morris et al. (2017) model places a transformed AR(1) prior on three tail parameters $\mathbf{w}_{tk}$, z_{tk}, σ_{tk} at each spatial knot k. The AR(2) extension replaces each AR(1) with a stationary AR(2). To preserve the skew-t marginal at every time point (the structural property the paper hinges on), the variance of every transformed parameter must equal one for all t, which by Yule–Walker fixes

$$\gamma_0 = 1, \qquad \gamma_1 = \frac{\phi_1}{1-\phi_2}, \qquad \gamma_2 = \phi_1\gamma_1 + \phi_2, \qquad \sigma_\epsilon^2 = 1 - \phi_1\gamma_1 - \phi_2\gamma_2, \tag{1}$$

all functions of (ϕ_1, ϕ_2) only. The factorization of the joint AR(2) prior used by the MCMC is

$$p(Y_1) = N(0,1), \tag{2}$$
$$p(Y_2 \mid Y_1) = N\left(\gamma_1 Y_1,\ 1-\gamma_1^2\right), \tag{3}$$
$$p(Y_t \mid Y_{t-1}, Y_{t-2}) = N\left(\phi_1 Y_{t-1} + \phi_2 Y_{t-2},\ \sigma_\epsilon^2\right), \quad t \ge 3. \tag{4}$$

The current implementation in `code/R/ar2/` preserves (**??**) but deviates from (**??**)–(**??**) and detailed balance in three places. This document covers what is wrong (*why fix*), what was changed (*how fix*), and what the regression tests now show (*results*).

Summary of state. Fixes 2 and 3 have been applied and are guarded by automated tests. Fix 1 is deferred – the bias is $O(1/n_t)$ and dominated by Fix 2 in any realistic regime – but is documented inline so the next maintainer cannot miss it.

	Fix 1 ($t=2$ conditional)	**Fix 2** (lag-2 MH term)	**Fix 3** (ϕ Hastings)
Status	deferred	applied	applied
Bias scale	$O(1/n_t)$	$O(1)$ on lag-2 ACF	$O(1)$ on ϕ posterior
Production sites	1	4	1
Tests	–	3	1

2 Fix 3: Hastings adjustment in `updatePhiAR2TS`

2.1 Why fix

The original `updatePhiAR2TS` proposes $\phi_j^{\text{can}} \sim N(\phi_j, \tau_j^2)$ and silently rejects whenever the candidate falls outside the AR(2) stationarity region (**??**)[1]. Silent rejection is operationally equivalent to a normal proposal *truncated* to the conditional stationarity interval

$$S_1(\phi_2) = (\phi_2 - 1,\ 1 - \phi_2), \qquad S_2(\phi_1) = \big(|\phi_1| - 1,\ 1 - |\phi_1|\big),$$

whose normalising constant $P(S_j; m) = \Phi\big(\frac{b-m}{\tau_j}\big) - \Phi\big(\frac{a-m}{\tau_j}\big)$ depends on the proposal mean m. The proposal is therefore asymmetric in $(\phi_j, \phi_j^{\text{can}})$, and the Hastings correction $\log P(S_j; \phi_j) - \log P(S_j; \phi_j^{\text{can}})$ must enter the acceptance ratio. It did not. Because $P(S_j; m)$ is maximised at $m = 0$ and decreases toward the endpoints, the missing term biases the chain *toward the centroid of the stationarity triangle*, i.e. toward $\phi = 0$. The AR(1) routine `updatePhiTS` handles this correctly; the AR(2) routine was a depth-1 copy of it.

2.2 How fix

We add the asymmetry correction at both component updates of `updatePhiAR2TS`. For ϕ_1:

```
s1.lo <- phi2 - 1
s1.hi <- 1 - phi2
log.P.cur1 <- log(pnorm(s1.hi, phi1,     mh.phi1) -
                  pnorm(s1.lo, phi1,     mh.phi1))
# ... draw can.phi1, check stability, compute can.ll ...
log.P.can1 <- log(pnorm(s1.hi, can.phi1, mh.phi1) -
                  pnorm(s1.lo, can.phi1, mh.phi1))

R <- can.ll - cur.ll +
     dnorm(can.phi1, p.m.1, p.s.1, log = TRUE) -
     dnorm(phi1,     p.m.1, p.s.1, log = TRUE) +
     log.P.cur1 - log.P.can1   # Hastings adjustment
```

Mirror block for ϕ_2 uses $S_2(\phi_1)$. The patch is a pure addition; the silent-rejection on the candidate side is unchanged.

2.3 Results

`tests/test_fix3_phi_hastings.R` runs a paired MCMC: same data ($n_t = 40$, $n = 4$ chains, truth $(\phi_1, \phi_2) = (0.45, 0.50)$ at the edges of both S_j intervals), same RNG seed for proposals, twelve replicate datasets. The buggy reference is inlined verbatim so the test still detects a revert.

Averaged median bias	ϕ_1	ϕ_2
buggy (no Hastings adjustment)	-0.0112	$+0.0078$
patched (Fix 3 applied)	-0.0053	$+0.0001$
ratio $\lvert patched\rvert / \lvert buggy\rvert$	0.48	0.01

The buggy chain shows the expected shrink-toward-zero signature on ϕ_1 (truth $0.45 > 0$, median 0.439). The patched chain reduces $|bias|$ on both components and the test asserts a bias-reduction ratio ≤ 0.7, which is met by a comfortable margin on both axes.

[1] $|\phi_2| < 1$, $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$; see `check_ar2_stability`.

3 Fix 2: missing lag-2 term in the Y_t MH ratio

3.1 Why fix

Under (??), the value Y_t enters *three* factors of the joint prior:

$$\underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{Y_t \text{ is lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{Y_t \text{ is lag-2}}.$$

The AR(1) template only requires the first two factors because lag-2 has no effect under AR(1). All three update functions in the AR(2) port (`updateTauTS_AR2`, `updateZTS_AR2`, `updateKnotsTS_AR2`) inherit this two-factor pattern and silently drop the lag-2 factor. The chain therefore satisfies detailed balance for the modified target

$$\widetilde{\pi}(Y_t) \propto p(Y_t \mid Y_{t-1}, Y_{t-2})\, p(Y_{t+1} \mid Y_t, Y_{t-1}),$$

not the AR(2) prior. The empirical signature is large: the stationary lag-2 ACF of the buggy chain differs from γ_2 by an $O(1)$ amount, and the marginal variance of the chain is inflated.

3.2 How fix

For each of the four call sites we add a $t + 2$ correction parallel to the existing $t + 1$ block. In `updateZTS_AR2`:

```
# Fix 2: z.star[k, t] enters the t+2 conditional as lag-2
if ((t + 2) <= nt) {
    z.star.next2 <- z.star[k, t + 2]
    R <- R +
        ar2_conditional_log_density(
            z.star.next2, t + 2,
            z.star[k, t + 1], can.z.star[k],
            phi1 = phi1, phi2 = phi2, moments = ar2.moments) -
        ar2_conditional_log_density(
            z.star.next2, t + 2,
            z.star[k, t + 1], z.star[k, t],
            phi1 = phi1, phi2 = phi2, moments = ar2.moments)
}
```

Edge case at $t = 1$: the companion $t + 1$ block evaluates $p(Y_2 \mid Y_1)$ via the $t == 2$ branch of `get_ar2_conditional_params`, which still uses the AR(1)-style form $N(\phi_1 Y_1, \sigma_\epsilon^2)$ instead of the stationary $N(\gamma_1 Y_1, 1 - \gamma_1^2)$ – the Fix 1 defect. The bias from this is $O(1/n_t)$ and dominated by Fix 2 itself, so Fix 1 is deferred; an inline `NOTE` comment at each of the four Fix 2 sites flags the dependency for future maintenance.

3.3 Results

Three layered tests cover this fix.

(a) Math: `test_fix2_lag2_mh.R`. A prior-only Y_t chain with the lag-2 correction toggled on or off. Truth $(\phi_1, \phi_2) = (0.7, -0.4)$, giving $\gamma_1 = 0.500$, $\gamma_2 = -0.050$.

Chain	mean	var	lag-1 ACF	lag-2 ACF
target	0	1.000	+0.500	−0.050
buggy (no $t + 2$ term)	+0.009	1.142	+0.608	+0.222
patched (Fix 2 applied)	+0.005	0.998	+0.498	−0.054

The buggy chain inflates the marginal variance by 14 % and gets the lag-2 autocorrelation *wrong in sign* (error 0.272, more than five times the tolerance). The patched chain matches the AR(2) target to four decimal places.

(b) Integration: `test_fix2_production.R`. Drives the real `updateZTS_AR2` with a zero-data configuration ($y = 0$, $\lambda = 0$, $x_t^\top \beta = 0$, $\tau = 1$, prec $= \mathbf{I}$), stubs out `updatePhiAR2TS` so ϕ stays at truth, and records z^* via the half-normal copula transform after each iteration. Empirical ACFs computed across iterations and time:

	lag-1 ACF error	lag-2 ACF error
patched (current code)	0.0002	0.0145
reverted (smoke check)	0.1121	0.2840

The reverted-code row was produced by temporarily commenting out the Fix 2 block in `updateZTS_AR2`; both errors blow past the 0.05 tolerance, confirming the test will fire on a future regression.

(c) Structural: `test_fix2_grep.R`. Asserts the literal pattern `(t + 2) <= nt` appears exactly four times in `update_params_ar2.R`, matching the four sites where Y_t enters another day's conditional density as lag-2. This is the cheapest layer and catches the broadest regression – "someone deleted a block" – across all three update functions, including `updateTauTS_AR2` and `updateKnotsTS_AR2` that lack their own integration tests.

4 Fix 1: stationary $t = 2$ conditional (deferred)

4.1 Why fix (eventually)

`get_ar2_conditional_params` returns, for $t = 2$, $N(\phi_1 Y_1, \sigma_\epsilon^2)$ instead of the stationary form (**??**). The simulator `simulate_ar2_standard` draws (Y_1, Y_2) jointly from the correct N_2, so the generator and MCMC density disagree on the joint of (Y_1, Y_2). The disagreement contributes a single density factor at $t = 2$ out of $n_t - 1$ transitions, so the integrated bias scales as $O(1/n_t)$ and is dominated by Fix 2 itself.

4.2 How fix (one-line patch)

```r
# AFTER (proposed)
if (t == 2) {
    g1    <- moments$gamma1
    sd.t2 <- sqrt(moments$gamma0 - g1^2)    # = sqrt(1 - gamma1^2)
    return(list(mean = g1 * lag1, sd = sd.t2))
}
```

The patched form reduces to the AR(1) case $N(\phi_1 Y_1, 1 - \phi_1^2)$ when $\phi_2 = 0$, since $\gamma_1 \to \phi_1$ in the limit.

4.3 Results

Not yet applied. The deferral is intentional: at $n_t \geq 30$ the bias is below the noise floor of any downstream summary we report, and the Fix 2 test infrastructure dominates the maintenance budget. The four Fix 2 sites each carry an inline `NOTE (Fix 1 dependency)` comment pointing back to this section so the issue is not silently forgotten.

5 Test layout and regression protection

All tests live in `code/R/ar2/tests/` and run via `Rscript test_*.R`. Each exits 0 on pass, 1 on fail.

Test file	Targets	Layer	Catches
`test_fix3_phi_hastings`	Fix 3	paired (real vs. buggy)	bias in ϕ posterior median
`test_fix2_lag2_mh`	Fix 2	math	lag-2 ACF distortion
`test_fix2_production`	Fix 2	integration	regression in `updateZTS_AR2`
`test_fix2_grep`	Fix 2	structural	deletion of any of the four $t+2$ blocks

Fix 3 has a single integration test because the bug is localised to one function. Fix 2 has three layered tests because the patch touches four call sites and only one of them is exercised end-to-end; the structural grep provides broad cheap coverage across the remaining three.

6 Resulting consistency

After Fixes 2 and 3, the AR(2) implementation matches the stationary AR(2) prior (**??**)–(**??**) for $t \geq 2$ at every site where the prior contributes – generator, MCMC density for Y_t, Metropolis–Hastings ratio for Y_t, and ϕ-update with the asymmetric-proposal correction. The remaining $t = 1$ discrepancy (Fix 1, $O(1/n_t)$) is documented in code. Each transformed time-varying parameter retains the $N(\mathbf{0}, \mathbf{I})$ marginal that the paper relies on to preserve the skew-t structure at each time point, so the structural guarantee of Morris et al. (2017) carries over to the AR(2) extension as intended.